

THE DIVERGENCE THEOREM OF GAUSS states that if V is the volume bounded by a closed surface S and $\mathbf{A}$ is a vector function of position with continuous derivatives, then

$$\iiint_V \nabla \cdot \mathbf{A} \, dV = \iint_S \mathbf{A} \cdot \mathbf{n} \, dS = \oiint_S \mathbf{A} \cdot d\mathbf{S}$$

where $\mathbf{n}$ is the positive (outward drawn) normal to S .

STOKES' THEOREM states that if S is an open, two-sided surface bounded by a closed, non-intersecting curve C (simple closed curve) then if $\mathbf{A}$ has continuous derivatives

$$\oint_C \mathbf{A} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{A}) \cdot \mathbf{n} \, dS = \iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

where C is traversed in the positive direction. The direction of C is called *positive* if an observer, walking on the boundary of S in this direction, with his head pointing in the direction of the positive normal to S , has the surface on his left.

GREEN'S THEOREM IN THE PLANE. If R is a closed region of the xy plane bounded by a simple closed curve C and if M and N are continuous functions of x and y having continuous derivatives in R , then

$$\oint_C M \, dx + N \, dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx \, dy$$

where C is traversed in the positive (counterclockwise) direction. Unless otherwise stated we shall always assume $\oint$ to mean that the integral is described in the positive sense.

Green's theorem in the plane is a special case of Stokes' theorem (see Problem 4). Also, it is of interest to notice that Gauss' divergence theorem is a generalization of Green's theorem in the plane where the (plane) region R and its closed boundary (curve) C are replaced by a (space) region V and its closed boundary (surface) S . For this reason the divergence theorem is often called *Green's theorem in space* (see Problem 4).

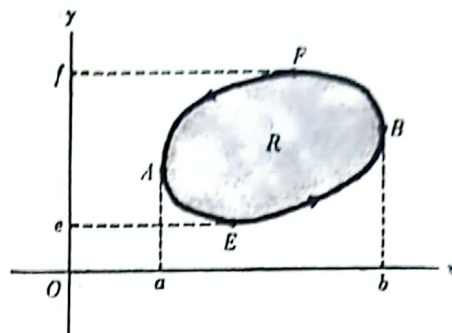
Green's theorem in the plane also holds for regions bounded by a finite number of simple closed curves which do not intersect (see Problems 10 and 11).

SOLVED PROBLEMS

GREEN'S THEOREM IN THE PLANE

1. Prove Green's theorem in the plane if C is a closed curve which has the property that any straight line parallel to the coordinate axes cuts C in at most two points.

Let the equations of the curves AEB and AFB (see adjoining figure) be $y=Y_1(x)$ and $y=Y_2(x)$ respectively. If R is the region bounded by C , we have



$$\begin{aligned} \iint_R \frac{\partial M}{\partial y} dx dy &= \int_{x=a}^b \left[\int_{y=Y_1(x)}^{Y_2(x)} \frac{\partial M}{\partial y} dy \right] dx = \int_{x=a}^b M(x, y) \Big|_{y=Y_1(x)}^{Y_2(x)} dx = \int_a^b [M(x, Y_2) - M(x, Y_1)] dx \\ &= - \int_a^b M(x, Y_1) dx - \int_b^a M(x, Y_2) dx = - \oint_C M dx \end{aligned}$$

Then

$$(1) \quad \oint_C M dx = - \iint_R \frac{\partial M}{\partial y} dx dy$$

Similarly let the equations of curves $EA F$ and $EB F$ be $x=X_1(y)$ and $x=X_2(y)$ respectively. Then

$$\begin{aligned} \iint_R \frac{\partial N}{\partial x} dx dy &= \int_{y=e}^f \left[\int_{x=X_1(y)}^{X_2(y)} \frac{\partial N}{\partial x} dx \right] dy = \int_e^f [N(X_2, y) - N(X_1, y)] dy \\ &= \int_f^e N(X_1, y) dy + \int_e^f N(X_2, y) dy = \oint_C N dy \end{aligned}$$

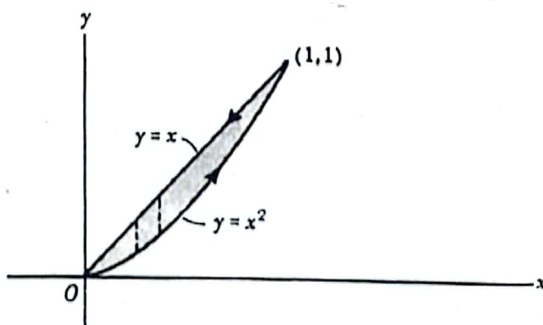
Then

$$(2) \quad \oint_C N dy = \iint_R \frac{\partial N}{\partial x} dx dy$$

Adding (1) and (2),
$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy.$$

- ★ 2. Verify Green's theorem in the plane for $\oint_C (xy + y^2) dx + x^2 dy$ where C is the closed curve of the region bounded by $y=x$ and $y=x^2$.

$y=x$ and $y=x^2$ intersect at $(0,0)$ and $(1,1)$. The positive direction in traversing C is as shown in the adjacent diagram.



Along $y = x^2$, the line integral equals

$$\int_0^1 ((x)(x^2) + x^4) dx + (x^2)(2x) dx = \int_0^1 (3x^3 + x^4) dx = \frac{19}{20}$$

Along $y = x$ from (1,1) to (0,0) the line integral equals

$$\int_1^0 ((x)(x) + x^2) dx + x^2 dx = \int_1^0 3x^2 dx = -1$$

Then the required line integral $= \frac{19}{20} - 1 = -\frac{1}{20}$.

$$\begin{aligned} \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy &= \iint_R \left[\frac{\partial}{\partial x}(x^2) - \frac{\partial}{\partial y}(xy + y^2) \right] dx dy \\ &= \iint_R (x - 2y) dx dy = \int_{x=0}^1 \int_{y=x^2}^x (x - 2y) dy dx \\ &= \int_0^1 \left[\int_{x^2}^x (x - 2y) dy \right] dx = \int_0^1 (xy - y^2) \Big|_{x^2}^x dx \\ &= \int_0^1 (x^4 - x^3) dx = -\frac{1}{20} \end{aligned}$$

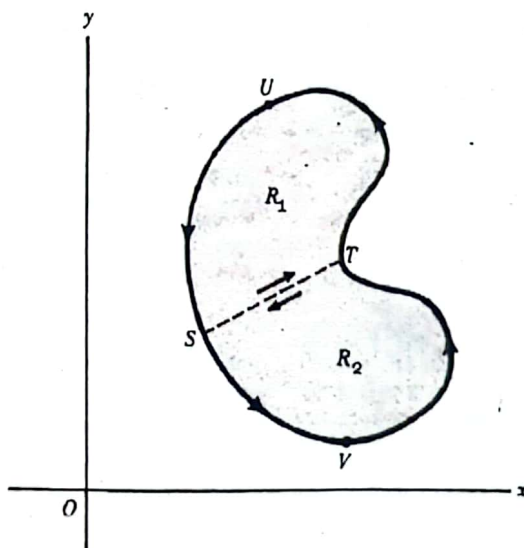
so that the theorem is verified.

3. Extend the proof of Green's theorem in the plane given in Problem 1 to the curves C for which lines parallel to the coordinate axes may cut C in more than two points.

Consider a closed curve C such as shown in the adjoining figure, in which lines parallel to the axes may meet C in more than two points. By constructing line ST the region is divided into two regions R_1 and R_2 which are of the type considered in Problem 1 and for which Green's theorem applies, i.e.,

$$(1) \quad \int_{STUS} M dx + N dy = \iint_{R_1} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$(2) \quad \int_{SVTS} M dx + N dy = \iint_{R_2} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$



Adding the left hand sides of (1) and (2), we have, omitting the integrand $M dx + N dy$ in each case,

$$\int_{STUS} + \int_{SVTS} = \int_{ST} + \int_{TUS} + \int_{SVT} + \int_{TS} = \int_{TUS} + \int_{SVT} = \int_{TUSVT}$$

using the fact that $\int_{ST} = -\int_{TS}$

Adding the right hand sides of (1) and (2), omitting the integrand,

Generalization of this to the case where the differential arc length ds of a closed curve C is replaced by the differential of surface area dS of a closed surface S , and the corresponding plane region R enclosed by C is replaced by the volume V enclosed by S , leads to Gauss' divergence theorem or Green's theorem in space.

$$\iint_S \mathbf{N} \cdot \mathbf{n} \, dS = \iiint_V \nabla \cdot \mathbf{N} \, dV$$

5. Interpret physically the first result of Problem 4.

If $\mathbf{A}$ denotes the force field acting on a particle, then $\oint_C \mathbf{A} \cdot d\mathbf{r}$ is the work done in moving the particle around a closed path C and is determined by the value of $\nabla \times \mathbf{A}$. It follows in particular that if $\nabla \times \mathbf{A} = \mathbf{0}$ or equivalently if $\mathbf{A} = \nabla \phi$, then the integral around a closed path is zero. This amounts to saying that the work done in moving the particle from one point in the plane to another is independent of the path in the plane joining the points or that the force field is conservative. These results have already been demonstrated for force fields and curves in space (see Chapter 5).

Conversely, if the integral is independent of the path joining any two points of a region, i.e. if the integral around any closed path is zero, then $\nabla \times \mathbf{A} = \mathbf{0}$. In the plane, the condition $\nabla \times \mathbf{A} = \mathbf{0}$ is equivalent to the condition $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ where $\mathbf{A} = M\mathbf{i} + N\mathbf{j}$.

6. Evaluate $\int_{(0,0)}^{(2,1)} (10x^4 - 2xy^3) dx - 3x^2y^2 dy$ along the path $x^4 - 6xy^3 = 4y^2$.

A direct evaluation is difficult. However, noting that $M = 10x^4 - 2xy^3$, $N = -3x^2y^2$ and $\frac{\partial M}{\partial y} = -6xy^2 = \frac{\partial N}{\partial x}$, it follows that the integral is independent of the path. Then we can use any path, for example the path consisting of straight line segments from $(0,0)$ to $(2,0)$ and then from $(2,0)$ to $(2,1)$.

Along the straight line path from $(0,0)$ to $(2,0)$, $y=0$, $dy=0$ and the integral equals $\int_{x=0}^2 10x^4 dx = 64$.

Along the straight line path from $(2,0)$ to $(2,1)$, $x=2$, $dx=0$ and the integral equals $\int_{y=0}^1 -12y^2 dy = -4$.

Then the required value of the line integral $= 64 - 4 = 60$.

Another Method.

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, $(10x^4 - 2xy^3) dx - 3x^2y^2 dy$ is an exact differential (of $2x^5 - x^2y^3$). Then

$$\int_{(0,0)}^{(2,1)} (10x^4 - 2xy^3) dx - 3x^2y^2 dy = \int_{(0,0)}^{(2,1)} d(2x^5 - x^2y^3) = 2x^5 - x^2y^3 \Big|_{(0,0)}^{(2,1)} = 60$$

7. Show that the area bounded by a simple closed curve C is given by $\frac{1}{2} \oint_C x dy - y dx$.

In Green's theorem, put $M = -y$, $N = x$. Then

$$\oint_C x dy - y dx = \iint_R \left(\frac{\partial}{\partial x}(x) - \frac{\partial}{\partial y}(-y) \right) dx dy = 2 \iint_R dx dy = 2A$$

where A is the required area. Thus $A = \frac{1}{2} \oint_C x dy - y dx$.

8. Find the area of the ellipse $x = a \cos \theta$, $y = b \sin \theta$.

$$\begin{aligned} \text{Area} &= \frac{1}{2} \oint_C x dy - y dx = \frac{1}{2} \int_0^{2\pi} (a \cos \theta)(b \cos \theta) d\theta - (b \sin \theta)(-a \sin \theta) d\theta \\ &= \frac{1}{2} \int_0^{2\pi} ab (\cos^2 \theta + \sin^2 \theta) d\theta = \frac{1}{2} \int_0^{2\pi} ab d\theta = \pi ab \end{aligned}$$

9. Evaluate $\oint_C (y - \sin x) dx + \cos x dy$, where C is the

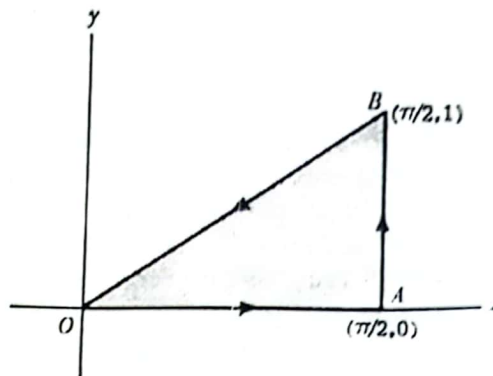
triangle of the adjoining figure:

(a) directly,

(b) by using Green's theorem in the plane.

(a) Along OA , $y = 0$, $dy = 0$ and the integral equals

$$\begin{aligned} \int_0^{\pi/2} (0 - \sin x) dx + (\cos x)(0) &= \int_0^{\pi/2} -\sin x dx \\ &= \cos x \Big|_0^{\pi/2} = -1 \end{aligned}$$



Along AB , $x = \frac{\pi}{2}$, $dx = 0$ and the integral equals

$$\int_0^1 (y - 1)0 + 0 dy = 0$$

Along BO , $y = \frac{2x}{\pi}$, $dy = \frac{2}{\pi} dx$ and the integral equals

$$\int_{\pi/2}^0 \left(\frac{2x}{\pi} - \sin x \right) dx + \frac{2}{\pi} \cos x dx = \left(\frac{x^2}{\pi} + \cos x + \frac{2}{\pi} \sin x \right) \Big|_{\pi/2}^0 = 1 - \frac{\pi}{4} - \frac{2}{\pi}$$

Then the integral along $C = -1 + 0 + 1 - \frac{\pi}{4} - \frac{2}{\pi} = -\frac{\pi}{4} - \frac{2}{\pi}$.

(b) $M = y - \sin x$, $N = \cos x$, $\frac{\partial N}{\partial x} = -\sin x$, $\frac{\partial M}{\partial y} = 1$ and

$$\begin{aligned} \oint_C M dx + N dy &= \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy = \iint_R (-\sin x - 1) dy dx \\ &= \int_{x=0}^{\pi/2} \left[\int_{y=0}^{2x/\pi} (-\sin x - 1) dy \right] dx = \int_{x=0}^{\pi/2} (-y \sin x - y) \Big|_0^{2x/\pi} dx \\ &= \int_0^{\pi/2} \left(-\frac{2x}{\pi} \sin x - \frac{2x}{\pi} \right) dx = -\frac{2}{\pi} (-x \cos x + \sin x) - \frac{x^2}{\pi} \Big|_0^{\pi/2} = -\frac{2}{\pi} - \frac{\pi}{4} \end{aligned}$$

in agreement with part (a).

Note that although there exist lines parallel to the coordinate axes (coincident with the coordinate axes in this case) which meet C in an infinite number of points, Green's theorem in the plane still holds. In general the theorem is valid when C is composed of a finite number of straight line segments.

10. Show that Green's theorem in the plane is also valid for a multiply-connected region R such as shown in the figure below.

The shaded region R , shown in the figure below, is multiply-connected since not every closed curve